

A decidable fragment of $\mathcal{ALCI}_{\text{RCC5}}$: the $\forall\text{PO}$ -free concepts

2026-07-16

Abstract

Concept satisfiability for $\mathcal{ALCI}_{\text{RCC5}}$ — the description logic $\mathcal{ALCI}$ over the five RCC5 base relations under abstract composition-table semantics — is open (Wessel 2002/2003), and reduces to a bounded-live-width property F6 whose status is the standing obstruction. We isolate a syntactically defined, unconditionally decidable fragment: concepts with *no* $\forall\text{PO}$ *subformula*. The key algebraic fact is that PO is the only relation whose recurrence along a vertical chain requires an all-phase safety condition not supplied by backward forcing or forward absorption; absent $\forall\text{PO}$, that condition is vacuous, the two-tier quotient is finite, and the live width is bounded by a computable function of the concept. The fragment retains $\exists\text{PO}$, $\forall\text{DR}$, $\exists\text{DR}$ and arbitrary vertical (PP/PPI) modalities. This note is a theorem-level write-up; its proof rests on the lemmas of the two-tier quotient development (`papers/two_tier_quotient.ALCIRCC5.tex`) and is not yet machine-certified. Its provenance is a cold research attack on the F6/W2' problem (GPT-5.5); we record it here with that status.

1 The fragment

We work in negation normal form (NNF) with inverse roles absorbed ($\exists\text{PP}^-.D = \exists\text{PPI}.D$), under strong-EQ semantics (EQ is identity; between distinct elements only $\{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$ occur), and abstract composition-table semantics: a model is a set with a total, converse-closed, composition-closed labelling of ordered pairs by RCC5 relations. Let $\text{cl}(C)$ be the closure of C (subformulas closed under the usual Fischer–Ladner conditions) and $m = |\text{cl}(C)|$.

Definition 1 ($\forall\text{PO}$ -free). A concept C is $\forall\text{PO}$ -free if no subformula of C has the form $\forall\text{PO}.D$.

The fragment is strictly weaker than banning “horizontal universals under horizontal existentials”: it still allows $\exists\text{PO}.D$, $\forall\text{DR}.D$, $\exists\text{DR}.D$, and unbounded vertical chains, so it retains infinite models and genuine horizontal existentials.

2 Why PO is the unstable case

Recall the four determining (singleton) compositions of proper relations,

$$\text{comp}(\text{DR}, \text{PPI}) = \{\text{DR}\}, \quad \text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\},$$

$$\text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}, \quad \text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\},$$

and the two relevant non-singleton facts

$$\text{comp}(\text{PP}, \text{PO}) = \{\text{DR}, \text{PO}, \text{PP}\}, \quad \text{comp}(\text{PPI}, \text{PO}) = \{\text{PO}, \text{PPI}\}.$$

Along a vertical chain $d_0 \text{ PP } d_1 \text{ PP } d_2 \dots$, the relation of an external witness e to the chain *stabilizes*: the RCC5 transition table admits only finitely many monotone exits among DR, PO, PP, PPI (external-relation stabilization). For the three “stable” relations this gives an all-phase witness by a determining composition:

- a DR-demand is carried to all deeper phases by $\text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$ (backward forcing);
- a PP-demand by $\text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$;
- a PPI-demand to all shallower phases by $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$ (forward absorption).

PO has *neither*: $\text{comp}(\text{PP}, \text{PO})$ and $\text{comp}(\text{PPI}, \text{PO})$ are non-singletons, so a witness that is PO-related at one phase need not be PO-related at another. This is precisely the point at which the two-tier quotient’s PO-coherence condition can be violated — but only by a *universal* PO-obligation.

Lemma 2 (Automatic PO-coherence). *Every $\forall\text{PO}$ -free concept is PO-coherent in the period-descriptor sense: for every recurrent descriptor $\text{Desc} = (\tau_1, \dots, \tau_p)$ and every $\exists\text{PO}.D \in \bigcup_j \tau_j$, any Hintikka type σ with $D \in \sigma$ is PO-safe for all phases τ_j with respect to PO-universal obligations.*

Proof. The type-safety test for a PO-edge $\tau_j \rightarrow \sigma$ requires, for the PO-universal part, that if $\forall\text{PO}.E \in \tau_j$ then $E \in \sigma$, and (since PO is symmetric) the analogous condition from σ to τ_j . In a $\forall\text{PO}$ -free concept no type contains any $\forall\text{PO}.E$, so the PO-universal part of $\text{Safe}(\tau_j, \sigma)$ is empty for every phase. The remaining requirements are the propositional/type conditions already secured by $D \in \sigma$; hence the same σ is safe for all phases. $\square$

3 Bounded quotient and decidability

Theorem 3 (Bounded live width; decidability). *For every satisfiable $\forall\text{PO}$ -free concept C there is a split-forest presentation of a model of C with live width at most*

$$K(C) = (1 + m)^2 2^{2^m}, \quad m = |\text{cl}(C)|.$$

Consequently concept satisfiability for the $\forall\text{PO}$ -free fragment of $\mathcal{ALCI}_{\text{RCC5}}$ is decidable.

Proof sketch. Start from any model of C . Along each infinite PP-chain record the recurrent set of Hintikka types with its cyclic order as a *period descriptor*; there are at most 2^m types and hence at most $D(C) \leq 2^{2^m}$ descriptors. By external-relation stabilization, DR/PP-demands obtain all-phase witnesses through backward forcing, PPI-demands through forward absorption, and (Lemma 2) PO-demands obtain all-phase safety because there are no $\forall\text{PO}$ constraints. Thus every recurrent descriptor has finitely many all-phase witnesses, at most one regular/designated witness per existential demand per descriptor, giving a two-tier quotient of size $Q(C) \leq D(C)(1+m) \leq (1+m) 2^{2^m}$.

The quotient network is checked by ordinary RCC5 path-consistency, which the patchwork property (Renz–Nebel) turns into global realizability; unfolding each kernel as its periodic PP-chain yields a model of C . All repeated vertical material is folded into the kernel loop, and cross-edges to external witnesses are constant quotient edges or vertical shadows, so the live interface width is bounded by the quotient size times the local number of existential demands: $K(C) \leq Q(C)(1+m)$. Decidability follows by enumerating all quotients up to this computable bound and checking the finite type, witness, safety, and RCC5 composition conditions. $\square$

Remark 4 (What the bound is and is not). The constant is not meant to be tight; only its *computability* matters, as decidability then follows from the conditional theorem of the width-barrier development (a computable live-width bound inhabits F6 for the fragment). The exponents reflect the type/descriptor counting, not a claimed complexity result.

Remark 5 (Relation to W2'). The mathematical fragment result uses an explicit quotient with designated witnesses; it does not require arbitrary glue steps to factor through a fixed coarse interface state, so it does not depend on the (false) coarse uniformization property W2'. A *catalogue* implementation with a fixed finite steering syntax would still need row-coloured steering or a principal-row-profile refinement, but the decidability of the fragment does not.

Remark 6 (Dependence on the two-tier Step 2, repaired 2026-08-28). The quotient above needs one relation per ordered pair of kernels, which the two-tier development supplies in Step 2 of its completeness theorem. That step originally defined it as a double limit $\lim_{i,j \rightarrow \infty} \rho(d_i, d_j)$, justified by external-relation stabilization “applied twice”. The inference is invalid — the stabilization lemma is *one-sided*, fixing an element and varying the chain, and the double limit need not exist — so the step was a genuine gap. This paper never states it: it uses stabilization one-sidedly, as the lemma allows. But it builds a two-tier quotient, so it inherits the step, and the reader should know where the repair is. It is in `papers/two_tier_quotient_ALCIRCC5.tex` (Remark “The lemma is one-sided” and Lemma “Simultaneous finite rectangles”), which replaces the limit by a constant finite *rectangle* — all that a quotient reading finitely many phases needs — and it is machine-checked in Lean as `fused_kq_all`. Theorem 3 and the bound $K(C)$ are unaffected.

4 Provenance and status

This fragment and its proof were produced by a cold research attack on the F6/W2' problem (GPT-5.5, on the packet `papers/cold_review_f6_w2prime`; the response and its supporting scripts are archived under `attack_req/`). The underlying ingredients — external-relation stabilization, backward-forcing safety for DR/PP, forward absorption for PPI, and the PO-coherence condition — are those of the two-tier quotient development (`papers/two_tier_quotient_ALCIRCC5.tex`), specialized here by a syntactic sufficient condition (Lemma 2). The result is *theorem-level and not machine-certified*; it should be read as a proof obligation whose lemmas are already present in that development, and it should be independently reviewed (and ideally Lean-transcribed) before being cited as established. Consistent with the project’s standing discipline, the honest label remains: *strongly supported, not certified*.

5 Relationship to the Lean artifact (added 2026-08-20)

Since this paper was written, a substantial Lean development for this same fragment has grown in `formal/POFreeLift.lean`. The relationship is *not* that the Lean transcribes the proof above, and the two must not be conflated when citing either. Precisely:

Shared engine. Every ingredient named in the sketch of Theorem 3 is machine-checked in the Lean file, under these names: external-relation stabilization (`external_stabilizes`), backward forcing for DR and PP (`backward_forcing_dr`, `backward_forcing_pp`), forward absorption for PPI (`forward_absorption_ppi`), and the $\forall$ PO-vacuity that makes Lemma 2 free (`mty_no_all_po`, resting on `pofree_cl_all`).

Different final step, and what is therefore NOT certified. This paper concludes with a *split-forest presentation* of live width at most $K(C) = (1 + m)^2 2^{2^m}$, and derives decidability from that bound. The Lean development instead builds a finite *certificate* (`MultiTier`: κ periodic kernels plus β externals) and obtains `Decidable` (`Satisfiable C0`) from a soundness theorem

(`multiTier_sound`) together with finite enumeration of certificate codes. Consequently the Lean file contains *no width statement at all*: neither $K(C)$, nor live width, nor the existence of a split-forest presentation, is machine-checked. A reader must not take the Lean development as certifying Theorem 3 as stated.

The split-forest idea is nonetheless load-bearing. The certificate’s shape *is* the split-forest discipline: kernels are the vertical PP/PPI skeleton (with a direction flag), externals carry the horizontal DR/PO completion, and the interface rows are the cross edges. So the decomposition remains the semantic foundation, exactly as the overview claims; what the Lean route dispenses with is the split forest as a formal object carrying a width measure, not the insight.

Consequence for a rewrite. Three registers are easy to conflate and should be separated explicitly in any revision: (i) the split-forest *insight*, which is alive and formalized in structure; (ii) the split-forest *presentation with its $K(C)$ bound*, which is this paper’s theorem and is a proof sketch, unreviewed and not machine-checked; (iii) the split-forest *plus patchwork bags and a parity automaton* of the full-logic argument (Theorems A/B/C), which is untouched by the Lean work and still gated on F6.